SwasthyaDrishti

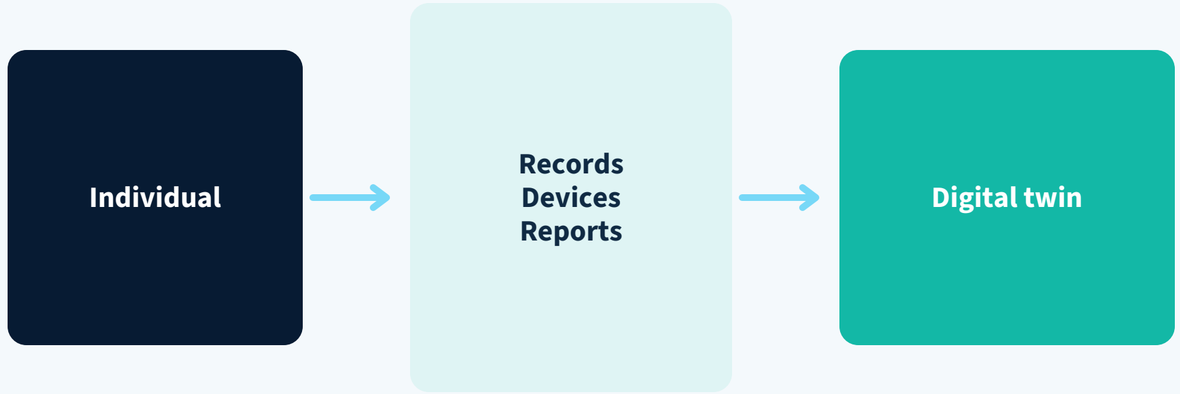
Digital Health Twin for Personalized Disease Monitoring



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A digital twin turns scattered health data into a coherent personal model

A digital health twin is a data-driven digital replica of an individual, built by integrating health observations over time.



Organize → analyze → monitor

Structured participant-level views support data-driven monitoring—not automated clinical decisions.

Health data is fragmented before it can become insight

Records

Devices

Reports

Longitudinal analysis becomes difficult when formats, coverage, and quality vary

Research prototype boundary: a structured monitoring framework—not a clinical deployment or diagnostic system.

Six objectives define the prototype

01

Integrate heterogeneous health data

02

Build participant-specific profiles

03

Apply preprocessing and pattern analysis

04

Explore early risk identification

05

Support personalized data insights

06

Address privacy, security & quality

From disconnected analysis to an integrated participant view

Conventional approach

Disconnected records

Limited participant context

Manual cross-checking

Hard-to-scale analysis

Proposed digital twin

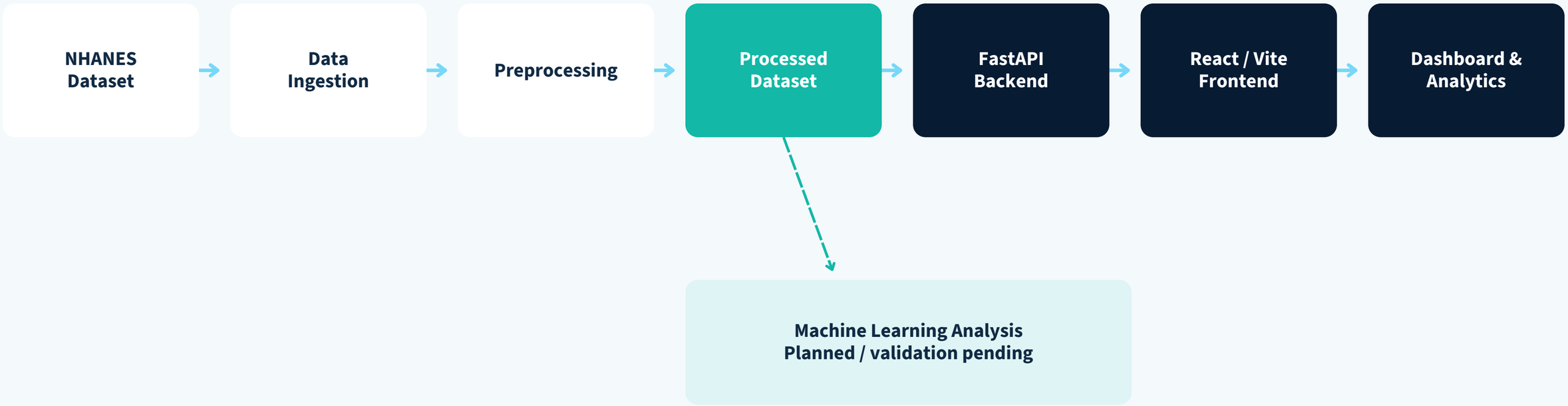
Centralized processed data

Participant-level profiles

API-enabled access

Scalable analytical foundation

A modular pipeline carries data from survey files to participant analytics



NHANES provides a rigorous public foundation

National Health and Nutrition Examination Survey

Public survey and examination data support participant-level exploration across demographics and measured health variables. No personally identifiable information is included.

Dataset audit before analysis

Core fields

Participant ID • Age • Gender

Body measurements

Sample size and variable count are cycle- and extract-specific; report values directly from the processed-data audit to keep statistics verifiable.

Preprocessing is a traceable, seven-step workflow



Duplicate checks • feature audits • missing-data handling

A focused stack supports rapid, modular delivery

Frontend

React + Vite

Backend

Python + FastAPI

Data processing

Python data-analysis libraries*

Dataset

NHANES

Deployment

Vercel + Render

Version control

Git + GitHub

*Use only libraries confirmed in the repository environment.

The prototype is deployed as a two-service web application




```
Frontend  https://frontend-dbgkntxs7-git-happens1.vercel.app/
Backend  https://swasthyadhristi-api.onrender.com/
Health  https://swasthyadhristi-api.onrender.com/api/health
```

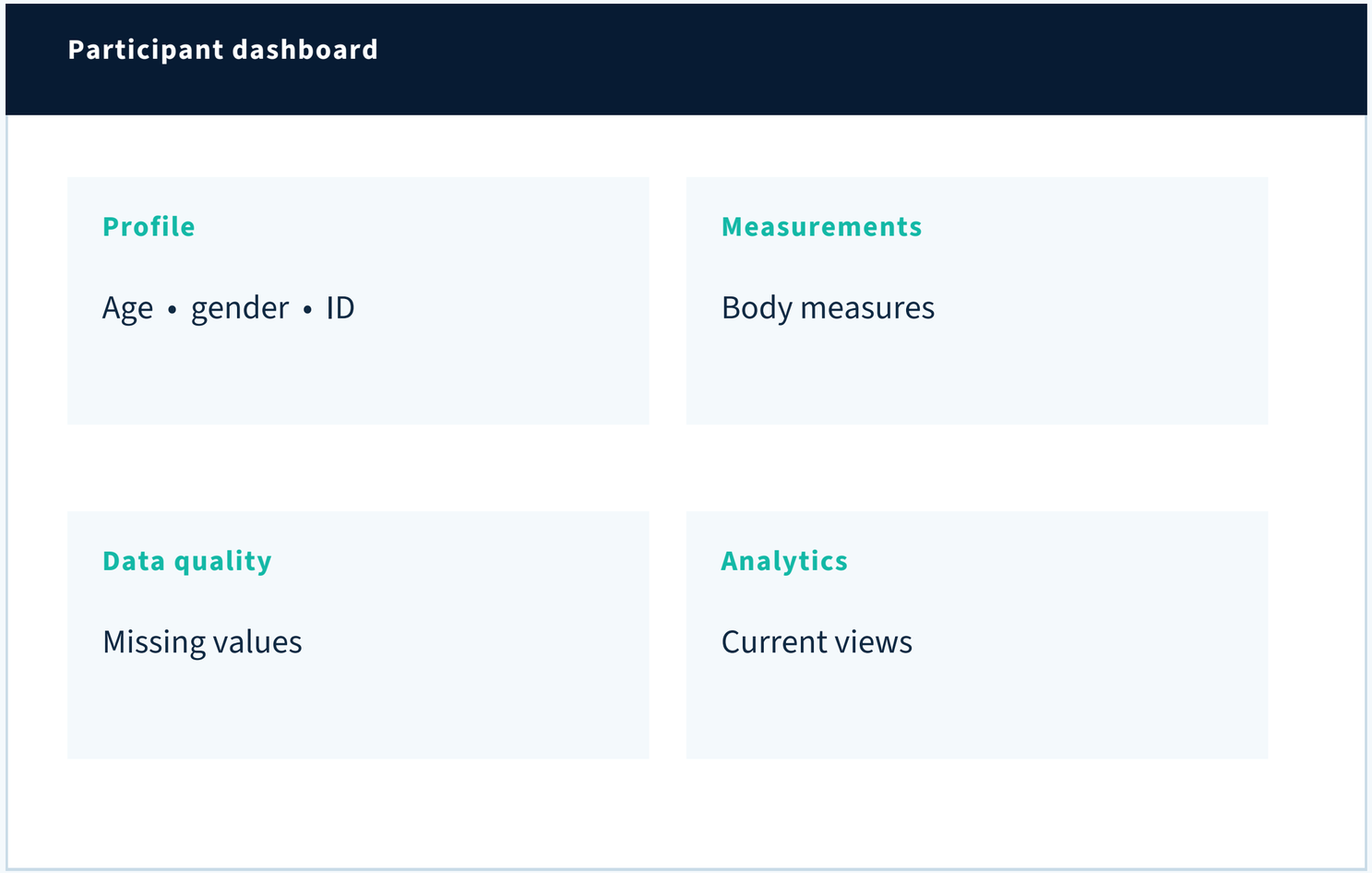
Current output: participant data, retrieved and visualized

Completed

- Deployed participant dashboard
- Demographic and measurement retrieval
- Frontend–backend API communication
- Missing-value views where implemented

Planned

Validated ML models and clinical interpretation



A credible prototype makes its limitations visible

Missing values
Survey fields are incomplete

Uneven measurements
Coverage varies by participant

Integration complexity
Cross-file joins need careful audits

Static survey scope
No continuous streaming data

Validation required
Not ready for clinical use

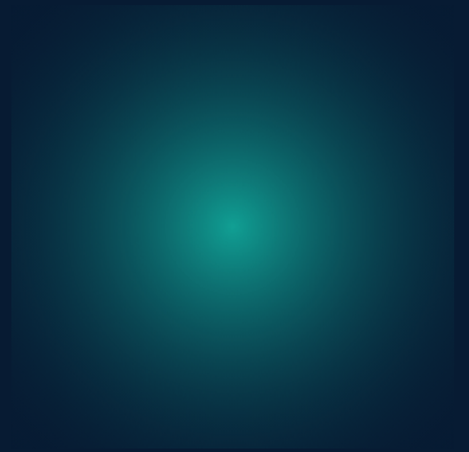
Not real-time monitoring
NHANES is cross-sectional / periodic

Future work moves from static profiles toward validated monitoring

1. Wearable + longitudinal data
2. Richer personal trend views
3. Validated ML models
4. Explainable risk indicators
5. Stronger privacy & security
6. Expert clinical validation

Thank You

Questions & Answers



SwasthyaDrishti demonstrates a foundation for structured participant-level health data analysis—integrating preprocessing, an API backend, and a deployed frontend.

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